

PHYSICS

INVESTIGATORY PROJECT

**Solar-Powered Automatic Street Light
Controller Using an LDR**

Sivabhargav G.V.

INDEX

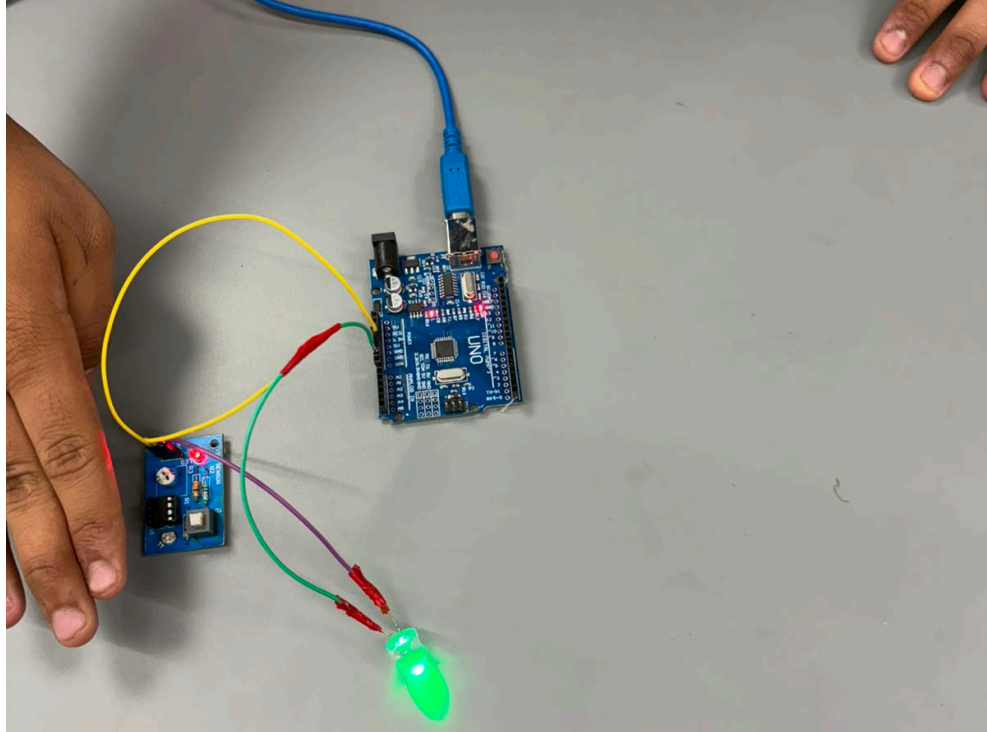
1. Aim / Objective
2. Apparatus / Components Required
3. Circuit Diagram
4. Theory / Underlying Scientific Principles
5. Working / Procedure
6. Observations
7. Result

1. Aim / Objective

To design, fabricate, and study a working model of a self-sustaining Automatic Street Light Controller that uses a solar photovoltaic panel to charge a battery during daylight hours, and that automatically switches ON a street light using a Light Dependent Resistor (LDR) circuit when ambient light falls below a calibrated threshold.

2. Apparatus / Components Required

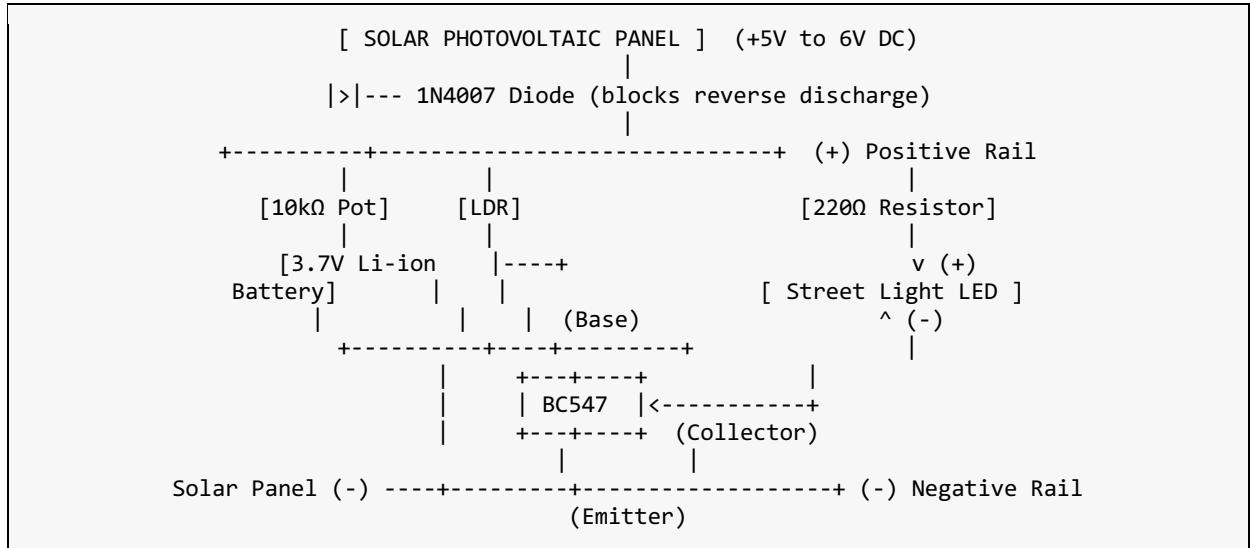
Component	Specification	Quantity
Solar Photovoltaic Panel	5V–6V DC, 100 mA, polycrystalline	1
Rechargeable Li-ion Battery	3.7V, 18650 cell, 1200–2200 mAh	1
Blocking Diode	1N4007 (1000V PIV, 1A forward current)	1
Light Dependent Resistor (LDR)	5 mm CdS photoresistor	1
NPN Transistor	BC547, $V_{BE(sat)} = 0.7V$, $I_{C(max)} = 100\text{ mA}$	1
Variable Potentiometer	10 k Ω / 50 k Ω rotary preset	1
Current-Limiting Resistor	220 Ω , carbon film	1
LED (Street Light)	High-efficiency white LED	1
Connecting Wires, Breadboard, Battery Holder	Standard laboratory grade	as required



Picture of the Apparatus: The LED can be seen glowing when hand is brought close to the apparatus.

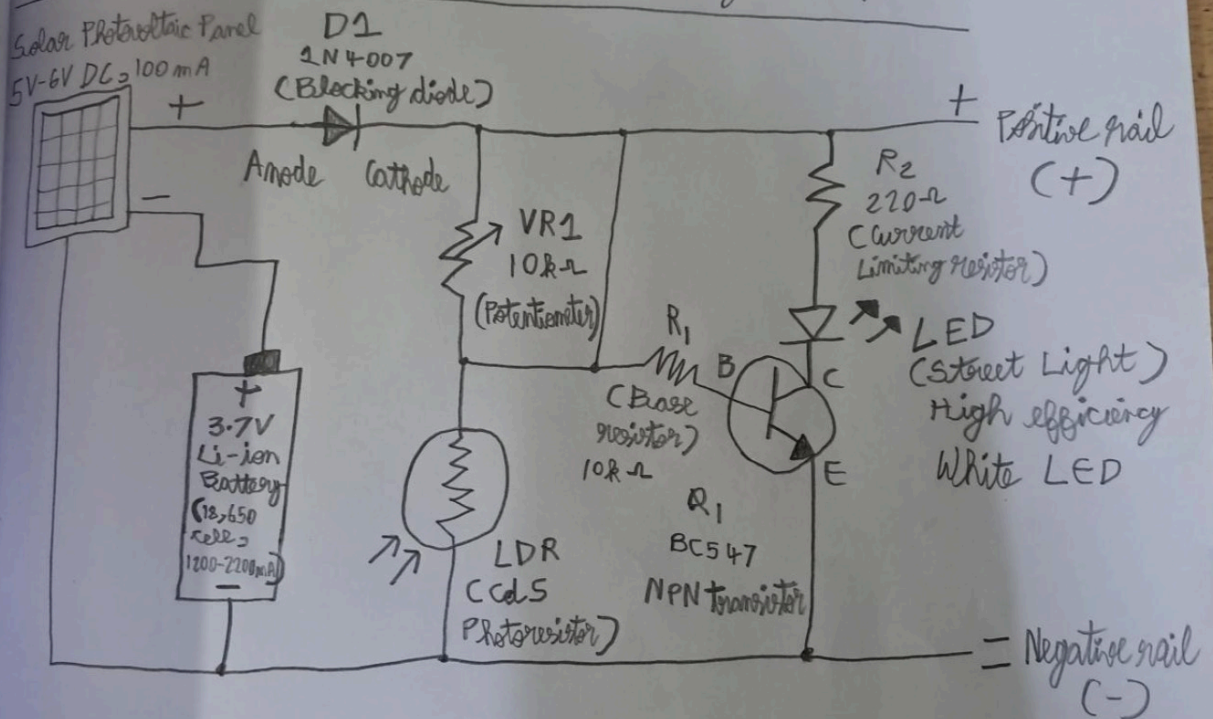
3. Circuit Diagram

The complete circuit integrates a solar charging stage with an LDR-based automatic switching stage, as shown below.



Hand-Drawn Circuit Diagram

Automatic Street Light Controller Using LDR ÷



→ LDR ÷ Light Dependent Resistor

→ LED ÷ Light Emitting Diode

4. Theory / Underlying Scientific Principles

The working of the model rests on three core physical principles.

A. The Photovoltaic Effect (Energy Generation)

When photons from sunlight strike the semiconductor surface of the solar panel, electrons are energised and dislodged from their atomic bonds. This flow of free electrons constitutes a direct current (DC), which is used to charge the rechargeable battery through a one-way protection diode.

B. Photoconductivity in Semiconductors (Ambient Light Sensing)

The LDR is fabricated from Cadmium Sulphide (CdS), a semiconductor whose electrical resistance depends on incident light intensity. When photons strike the CdS track, additional charge carriers are freed, so the resistance falls sharply in bright light. In darkness, the scarcity of free carriers causes the resistance to rise to the order of megaohms (M Ω).

C. Transistor Switching Action (Automation)

The BC547 NPN transistor acts as an electronic switch. The LDR and potentiometer together form a potential divider whose output sets the base voltage (V_{BE}) of the transistor. When V_{BE} exceeds the forward-bias threshold of about 0.7 V, the transistor is driven into saturation (closed switch), completing the circuit to the street light LED. When V_{BE} is below 0.7 V, the transistor remains in cut-off (open switch) and the LED stays OFF.

5. Working / Procedure

Construction

1. Connect the LDR and the potentiometer in series across the supply rails to form a voltage-divider network.
2. Connect the junction of the LDR and the potentiometer to the base terminal of the BC547 transistor.
3. Connect the emitter of the transistor to the negative rail, and the collector to the street light LED (in series with the $220\ \Omega$ current-limiting resistor) leading back to the positive rail.
4. Connect the solar panel to the battery through the 1N4007 diode, so that the panel charges the battery but current cannot flow backward into the panel at night.
5. Mount the LDR so that it faces ambient/sky light and is shielded from the street light LED itself, to prevent feedback oscillation.

Phase 1 — Daytime Cycle (Charging and Inaction)

- Sunlight generates a potential difference across the solar panel, which charges the Li-ion battery through the blocking diode.
- Bright ambient light keeps the LDR resistance very low, pulling the transistor's base voltage close to 0 V.
- Since V_{BE} is far below 0.7 V, the transistor stays in cut-off and the street light LED remains OFF.

Phase 2 — Night-time Cycle (Isolation and Automatic Lighting)

- As ambient light fades, the solar panel's output falls to 0 V, and the diode prevents the battery from discharging back into it.
- Darkness drives the LDR resistance up to several mega-ohms, forcing current through the potentiometer into the transistor's base.
- V_{BE} rises past 0.7 V, driving the transistor into saturation. The stored battery energy then flows through the $220\ \Omega$ resistor and switches the street light LED fully ON.

Testing

Cover the LDR with a hand or an opaque cap to simulate darkness — the street light LED should turn ON almost instantly. Shine a torch or smartphone flashlight on the LDR to simulate daylight — the LED should turn OFF. The potentiometer can be rotated to calibrate the light threshold at which switching occurs.

6. Observations

- The LDR circuit was tested under normal room-light conditions, with the LDR exposed to the surrounding light.
- Initially, the LED remained switched off when sufficient light was falling on the LDR.
- A hand was gradually brought closer to the LDR, thereby blocking an increasing amount of light from reaching its surface.
- As the intensity of light falling on the LDR decreased, the LED began to glow.
- When the hand was brought sufficiently close to the LDR, the LED remained illuminated.
- When the hand was moved away, allowing more light to fall on the LDR, the LED returned towards its previous state.
- The observation was repeated several times, and the same response was obtained each time.
- This demonstrated that the circuit responds to changes in the intensity of light falling on the LDR.
- The observation confirms the basic working principle of an automatic street-light system, in which the light is switched on when the surrounding illumination decreases.

7. Result

The Solar-Powered Automatic Street Light Controller was successfully fabricated and tested. The street light LED remained OFF under bright and moderate ambient light and switched ON automatically and consistently once illumination fell below the calibrated threshold, confirming that the LDR–transistor combination functions correctly as a light-activated automatic switch.